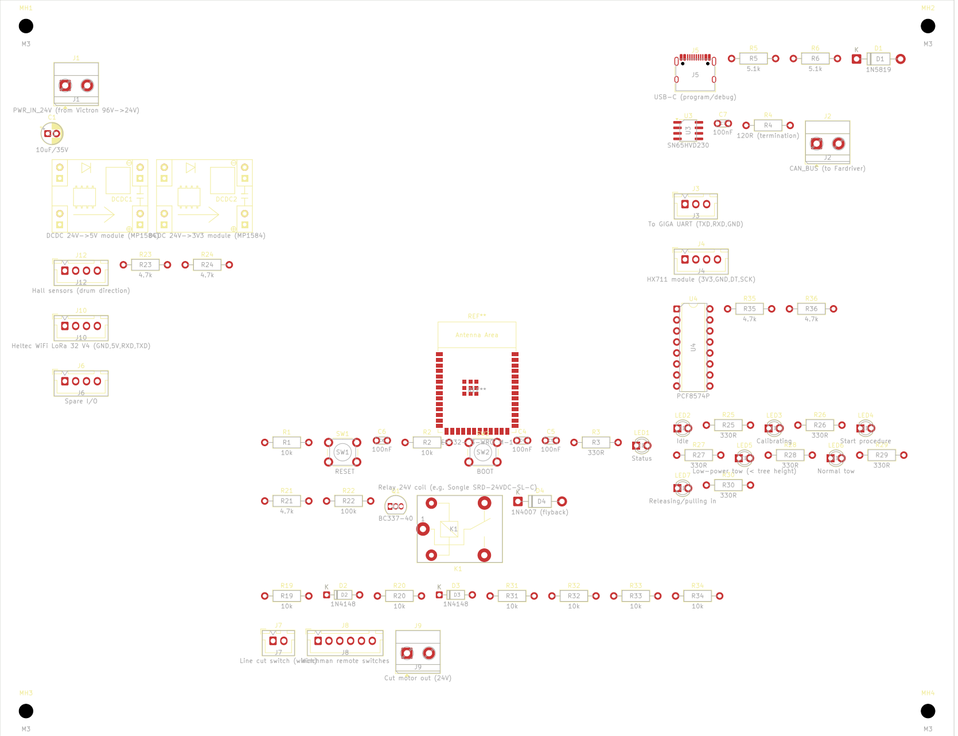
MH1
M3
MH2
M3
J1
PWR_IN_24V (from Victron 96V->24V)
C1
100uF/35V
DCDC1
DCDC2
DCDC 24V->5V module (MP1584)
DCDC 24V->3V3 module (MP1584)
J12
Hall sensors (drum direction)
R23
4.7k
R24
4.7k
J10
Heltec WIFI LoRa 32 V4 (GND,5V,RXD,TXD)
J6
Spare I/O
REF**
Antenna Area
R1
10k
SW1
RESET
C6
100nF
R2
10k
SW2
BOOT
C5
100nF
C4
100nF
R3
330R
LED1
Status
R21
4.7k
R22
100k
Q1
BC337-40
Relay 24V coil (e.g. Songle SRD-24VDC-SL-C)
D4
1N4007 (flyback)
K1
K1
R19
10k
D2
1N4148
R20
10k
D3
1N4148
R31
10k
R32
10k
R33
10k
R34
10k
J7
Line cut switch (watch)
J8
Human remote switches
J9
Cut motor out (24V)
J5
USB-C (program/debug)
R5
5.1k
R6
5.1k
D1
1N5819
U3
SN65HVD230
C3
100nF
R4
120R (termination)
J2
CAN_BUS (to Fardriver)
J3
To GIGA UART (TXD,RXD,GND)
J4
HX711 module (3V3,GND,DT,SCK)
U4
U4
PCF8574P
R35
4.7k
R36
4.7k
LED2
Idle
R25
330R
LED3
Calibrating
R26
330R
LED4
Start procedure
LED5
Releasing/pulling in
R27
330R
LED6
Low-power tow (< tree height)
R28
330R
LED7
Normal tow
R29
330R
R30
330R
MH3
M3
MH4
M3